



TK-TD modelling II

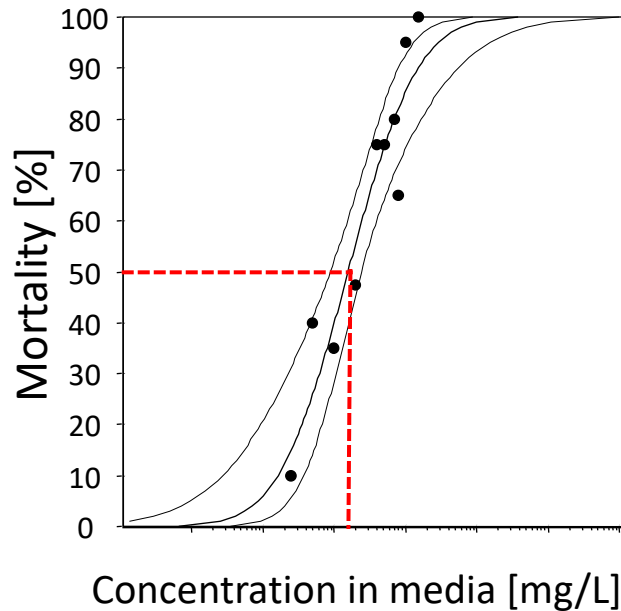
Survival Modelling



André Gergs
June 2026

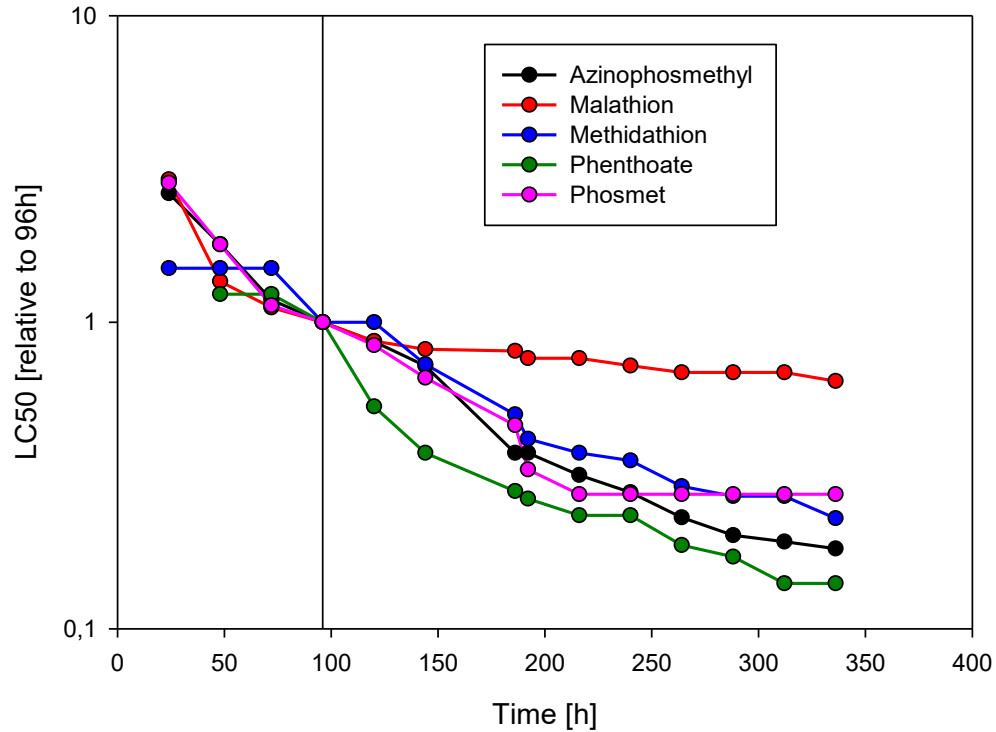


Lethal effects



- Effects (e.g. **% mortality**) are frequently determined at a certain **point in time** as a function of xenobiotic concentration in the media (**exposure concentration**)
- e.g. *Daphnia magna* acute toxicity test after 48h
- Logistic (or probit) regression allows the determination of the **LC₅₀** (or EC₅₀) value
- LC₅₀ is the **medium lethal concentration**, i.e. the concentration that is required to kill half of the population.

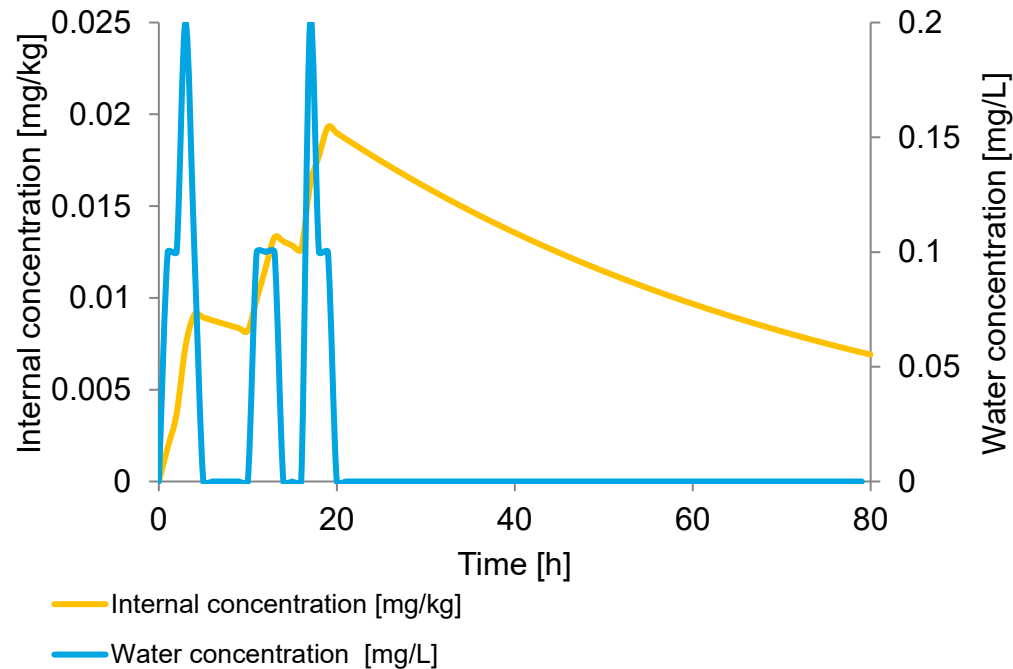
LC₅₀ depends on exposure duration



// LC₅₀ decreases with increasing exposure duration in guppies

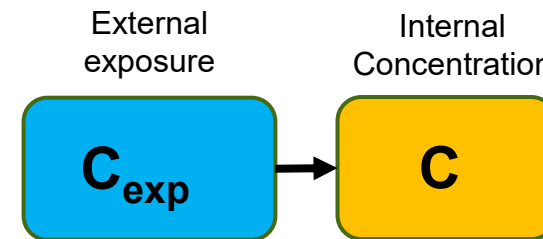
// Dose-response curve is a snap-shot of effects

Toxicokinetics



// When relating the effect to the external concentration, we assume that the chemical uptake and effects processes are instantaneous

// However, toxicokinetics may be a factor limiting toxicity in time



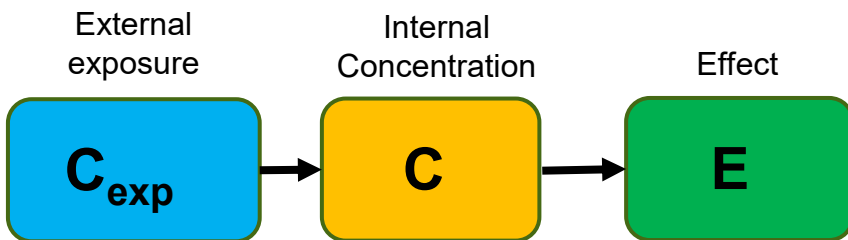
Processes leading to toxic effects

I. Exposure	II. Toxicokinetics	III. Toxicodynamics
degradation (environment)	uptake (in organism)	interaction of xenobiotic with target
distribution (environment)	distribution & metabolism (in organism)	mode of action
bioavailability	Excretion, elimination	


Bioaccumulation

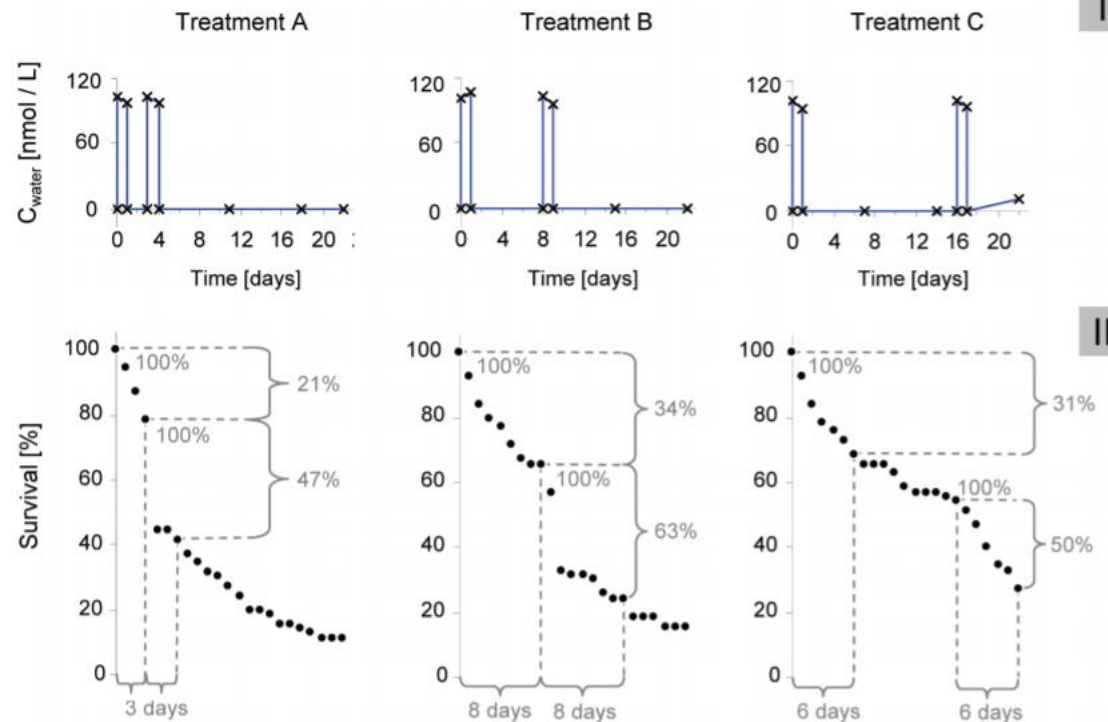
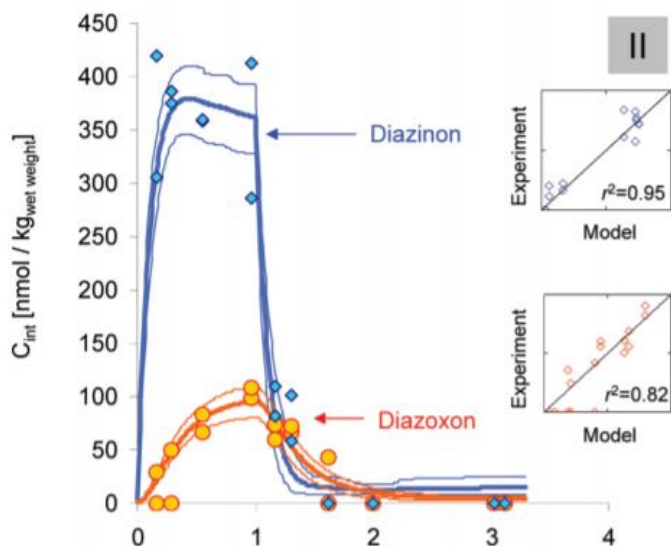

Toxicity

Does the internal concentration cause the effect?

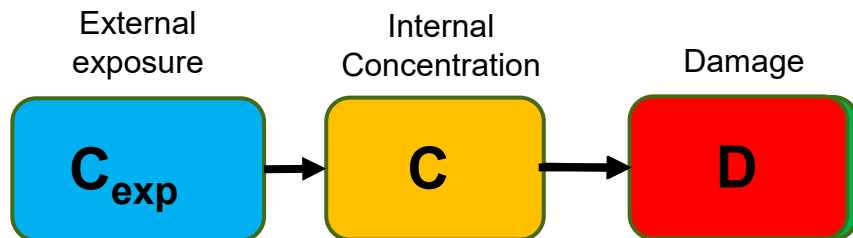


// Fast TK identified in the data and fitted with modelling.

// But deaths continued long after exposure ended



Linking toxicokinetics to effects

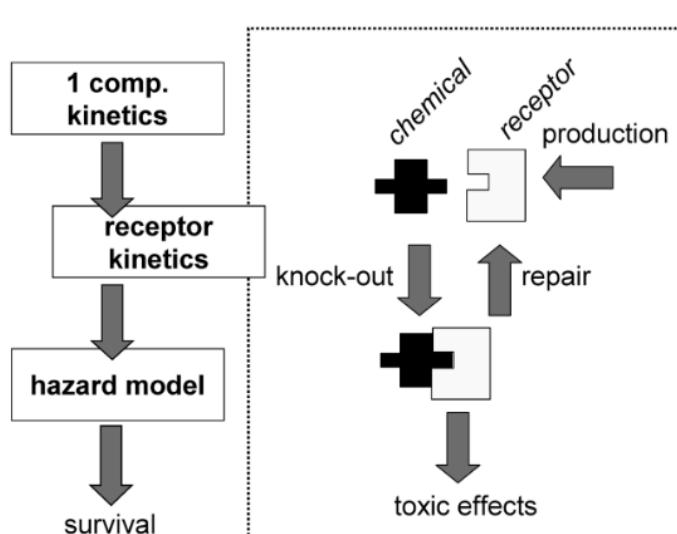


// When relating the effect to the internal concentration, we assume that the effects processes is instantaneous

// 'Damage' is introduced to represent the processes leading to an effect

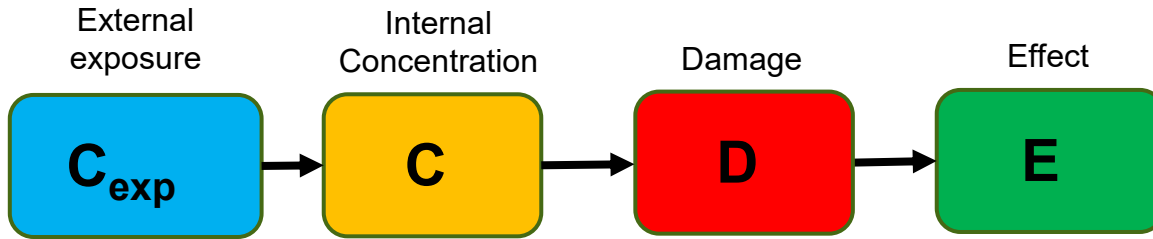
// Damage is abstract, fluxes are not necessarily mass fluxes

// Damage accrues proportional to the internal concentration, is eliminated proportional to damage.



Jager and Koojman <https://doi.org/10.1021/es050817y>

Scaling of TK and D - reduced model



$$\frac{dC_{int}}{dt} = k_u C_w - k_e C_{int} = k_e (K_{iw} C_w - C_{int})$$

$$C_{int}^* = \frac{C_{int}}{K_{iw}}$$

$$\frac{dC_{int}^*}{dt} = k_e (C_w - C_{int}^*)$$

$$\frac{dD_w}{dt} = k_e (C_w - D_w)$$

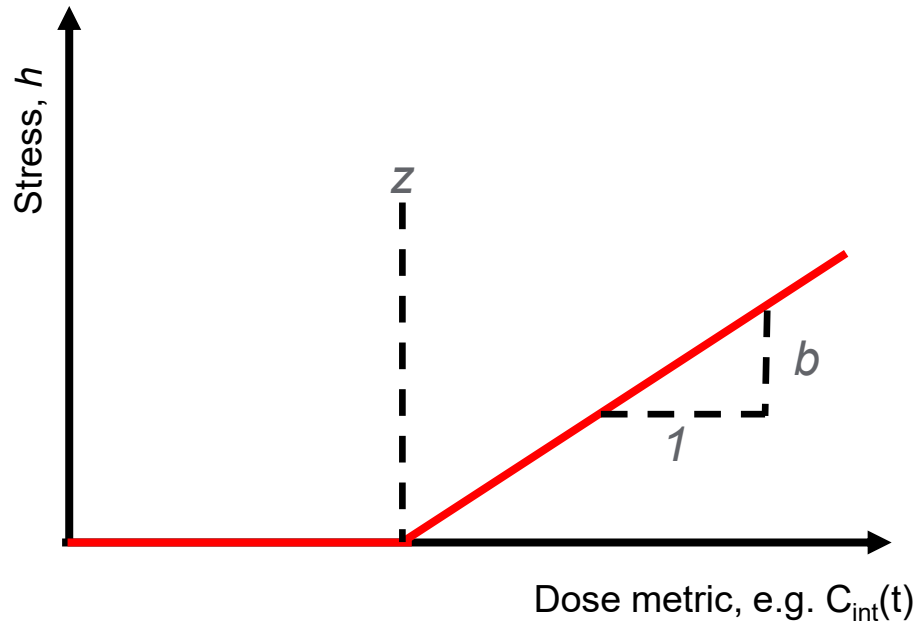
// When quantities cannot be measured, it is best to scale them (somehow).

// The scaled internal concentration has the same dimensions as external concentration

// If we assume $k_u/k_e = \text{BCF}_{\text{kinetic}} = 1$ the TK model can be reduced by one parameter

// 'Damage' can be scaled in a similar way

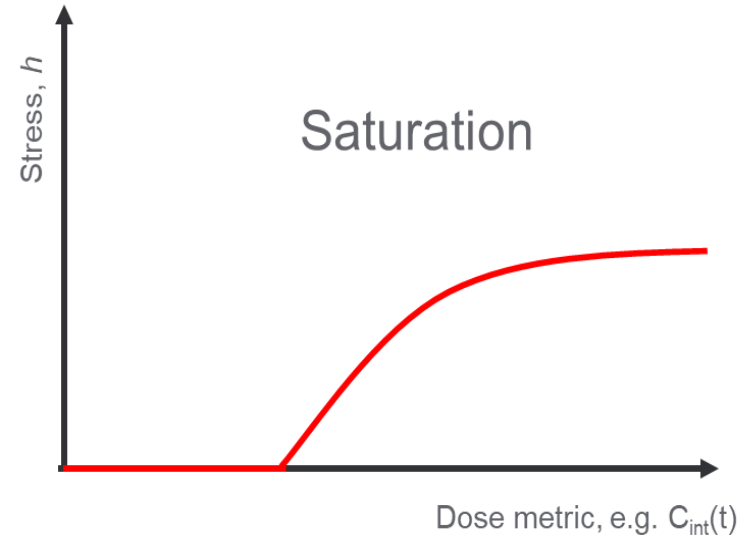
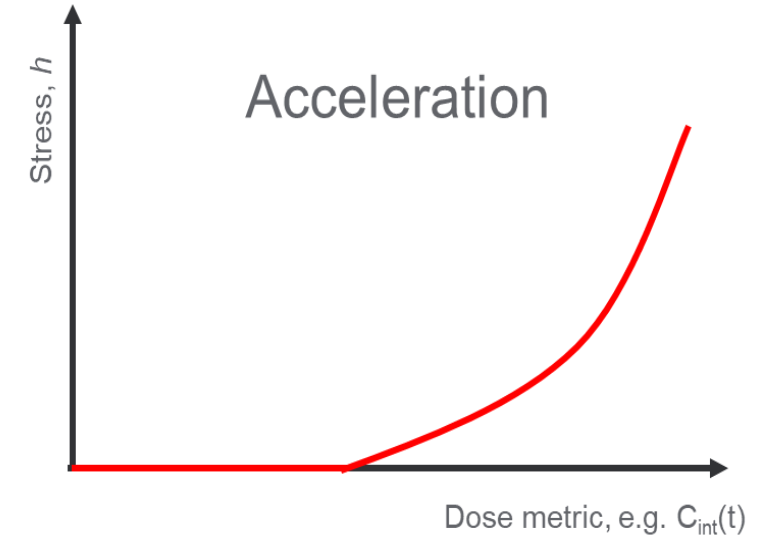
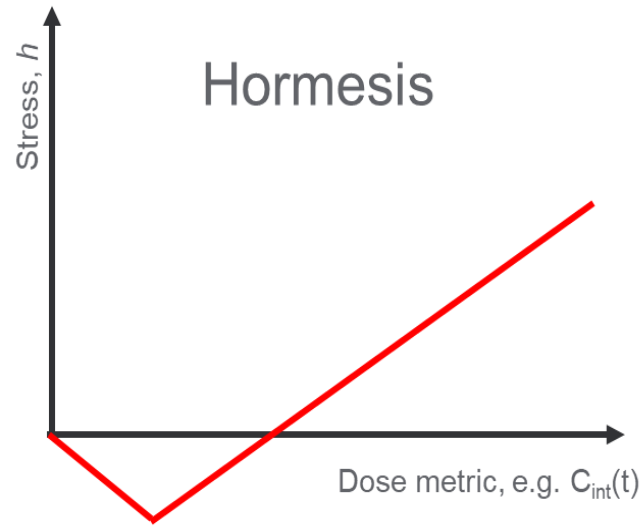
How to simulate effects?



- // Typically, TKTD models assume a linear-with-threshold response
 - // Some dose can be tolerated without effect
 - // Beyond the threshold (z), each unit of dose has the same additional increase in stress
- // Dose metric can be the external concentration, internal concentration (of parent or metabolite) or damage
- // Stress is reversible and acts on a state variable or parameter to alter the model behaviour.

$$h = b \max(\text{Dose} - z, 0)$$

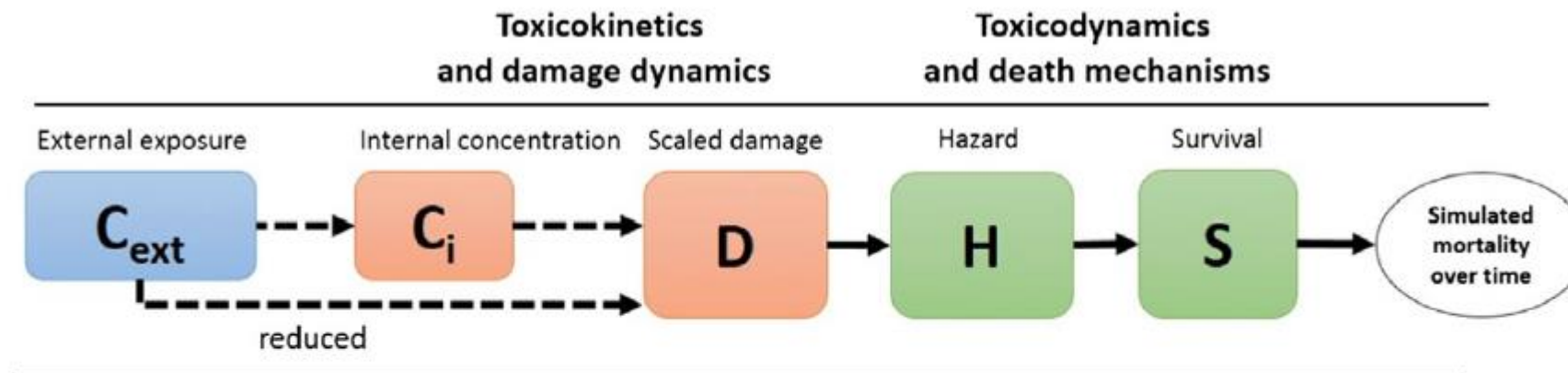
Other stress relationships



Figures by Neil Sherborne

Toxicokinetic-toxicodynamic (TK-TD) models

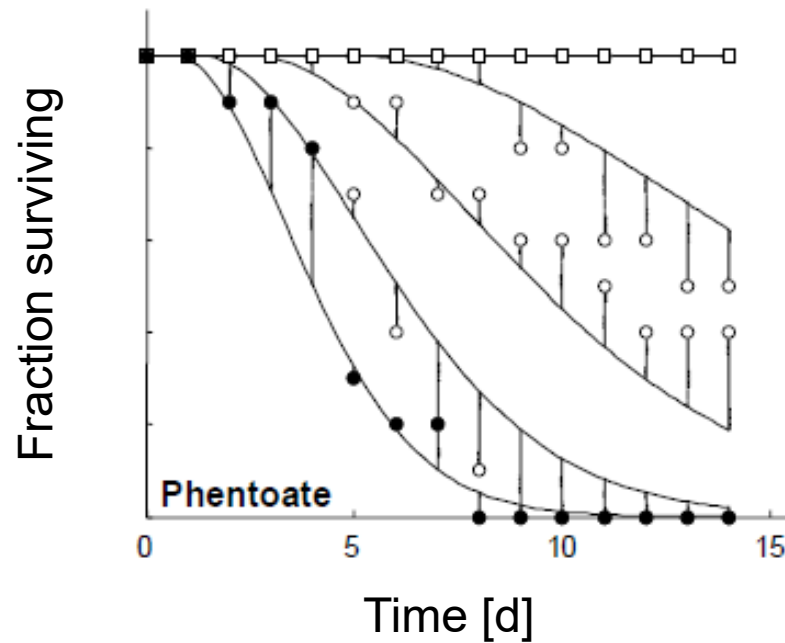
General Unified Threshold Model of Survival (GUTS)*



- D can depend either on
- External concentration via
 k_D : dominant rate constant
('scaled internal concentration' in Jager et al. (2011))
- or
- Internal concentration C_i via
 k_{in} : uptake rate constant
 k_{out} : depuration or elimination rate constant
 k_R : Damage recovery rate constant

*Jager et al. (2011): ES&T 45, 2529–2540

Application of TK-TD models



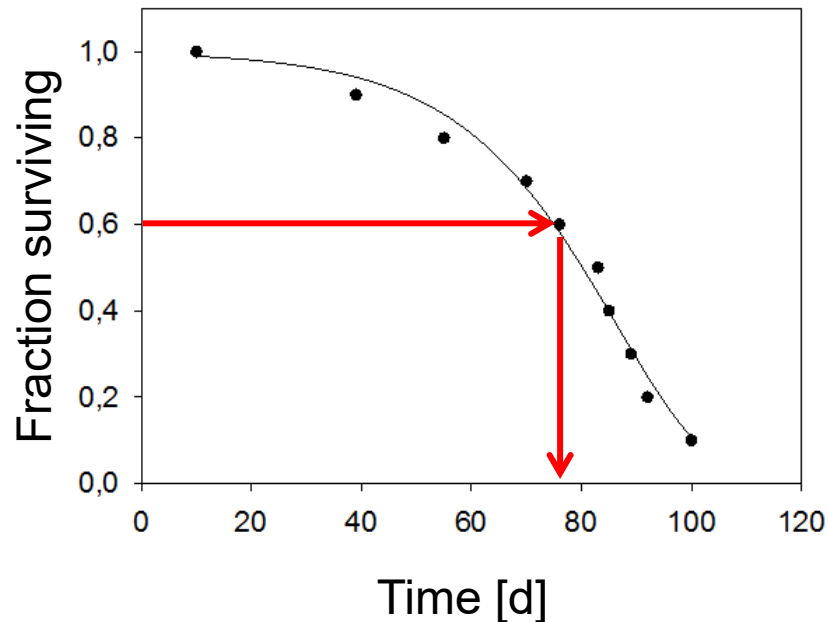
// **Understanding** rather than describing toxic effects

// **Process-based** modelling of the time course of toxic effects

// Applicable for **time variable exposure**

// **Prediction** of unknown scenarios

Distribution of survivorship within populations



// Mortality follows a **frequency distribution** within a population

// Why don't individuals die at the same time?

→ Individual differences in sensitivity
(**Individual tolerance**)

→ Death is a stochastic event
(**Stochastic death**)

// Both concepts are based on the idea that effects occur beyond a certain **threshold** (e.g. for C_B)

Individual tolerance



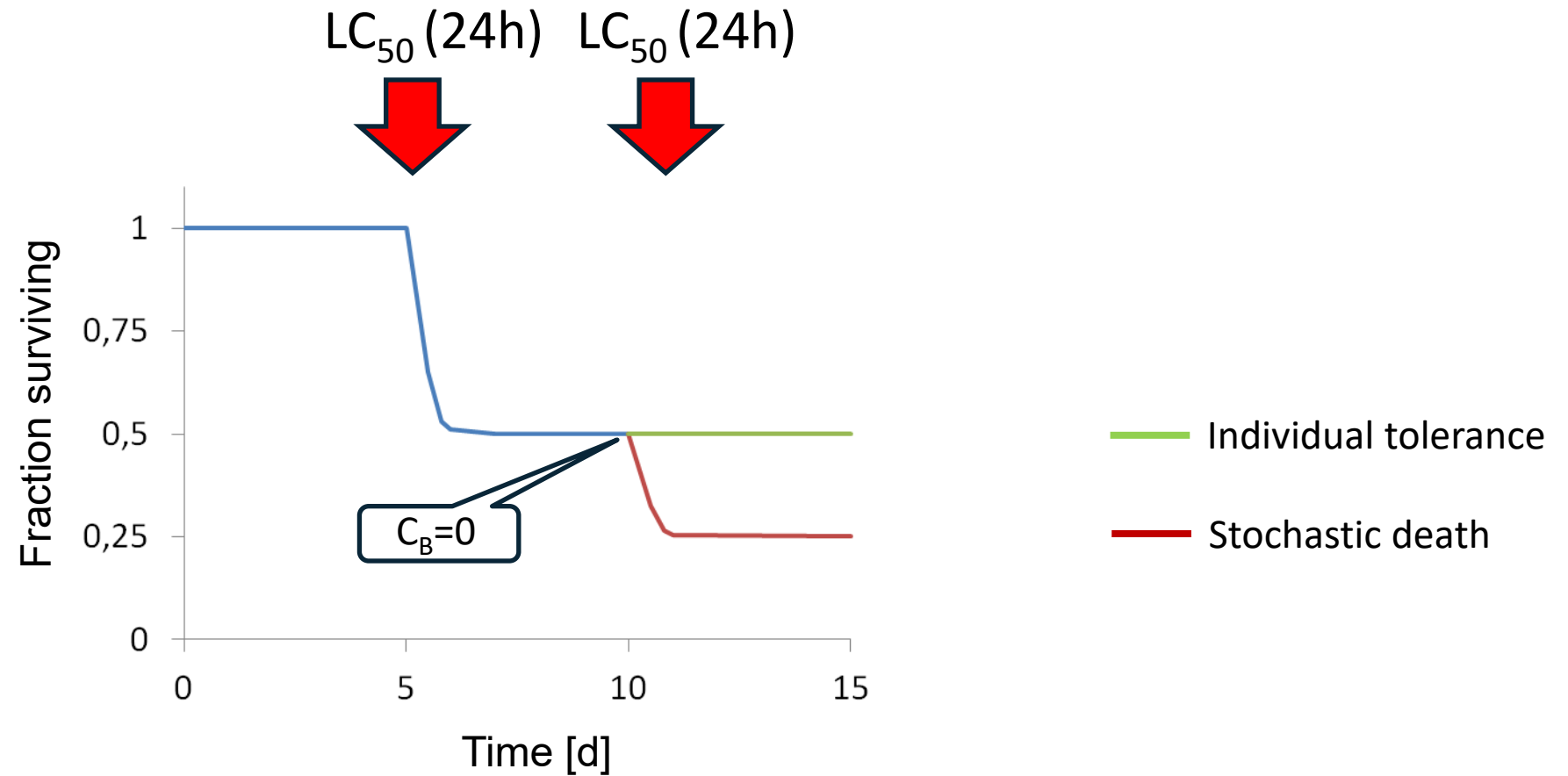
- // Threshold for effect is **different among individuals**
- // If threshold is exceeded effect is immediate (e.g. death: yes/no)
- // e.g. different tolerances for boring lectures
- // Distribution of the threshold within a population can be described by a **log-logistic frequency distribution function**
- // Survival probability (at time t) is calculated from the threshold function for an effect and the background mortality

Stochastic death



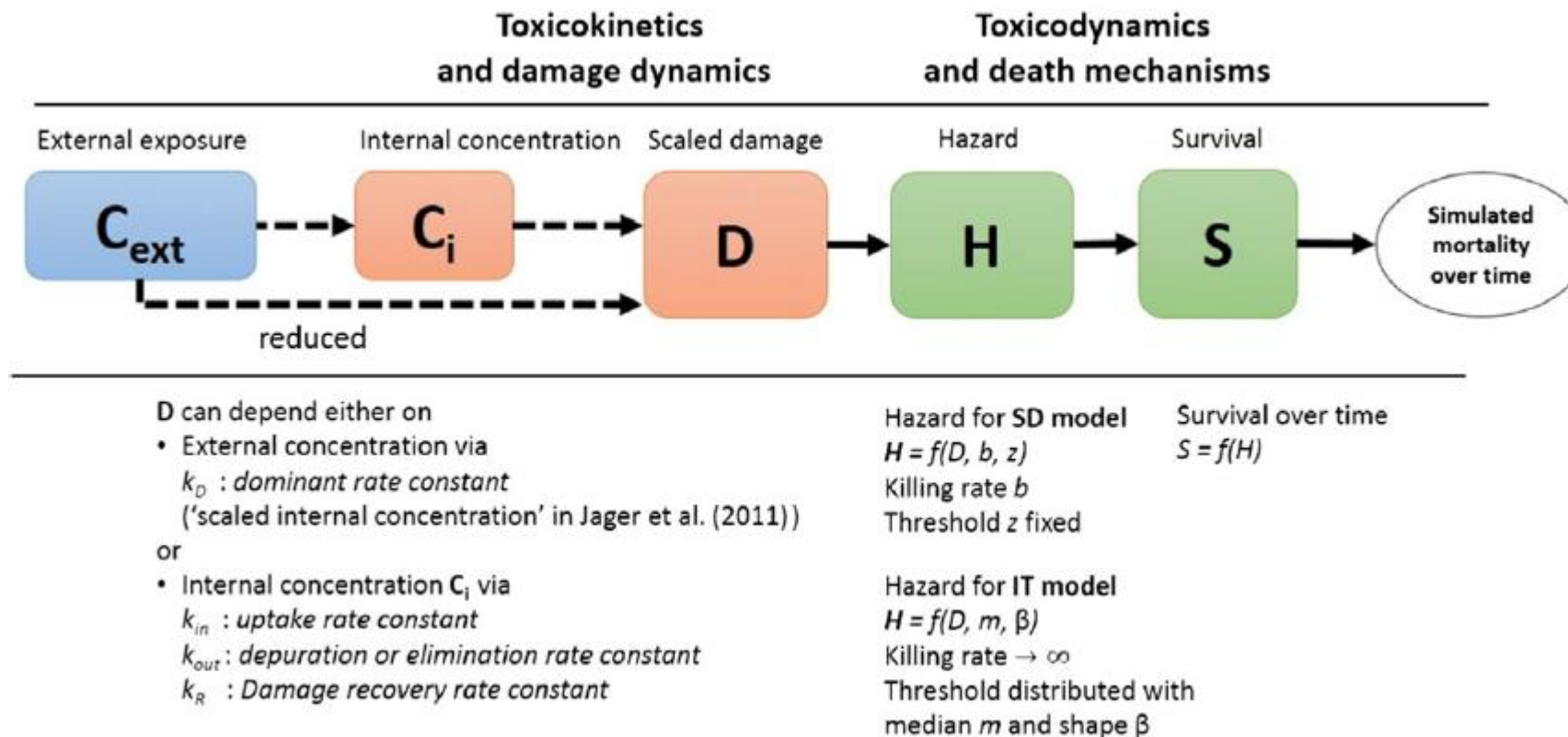
- // Threshold for effect is the **same for all individuals**
- // Hazard increases with concentration beyond the threshold
- // e.g. the chance of getting hit by a car is the same for all chicken when crossing a street
- // Hazard increases with number of street crossings
- // Survival probability is calculated based on the hazard rate

Stochastic death vs individual tolerance



Toxicokinetic-toxicodynamic (TK-TD) models

General Unified Threshold Model of Survival (GUTS)*



*Jager et al. (2011): ES&T 45, 2529–2540

GUTS equations at a glance

TOXICOKINETICS AND DAMAGE DYNAMICS	
Reduced GUTS (GUTS RED)	Full GUTS
No internal concentration in the model, scaled damage expressed in units of the external (water) concentration (D_w)	Internal concentration explicitly modelled, scaled damage expressed in units of the internal concentration (D_i)
$\frac{dD_w(t)}{dt} = k_D \times (C_w(t) - D_w(t))$	$\frac{dC_i(t)}{dt} = k_{in} \times C_w(t) - k_{out} \times C_i(t)$ $\frac{dD_i(t)}{dt} = k_R \times (C_i(t) - D_i(t))$
X	
TOXICODYNAMICS AND DEATH MECHANISM	
SD model	IT model
$\frac{dH(t)}{dt} = b \times \max(0, D(t) - z)$ $S_{SD}(t) = e^{-H(t)} \times e^{-h_b \times t}$	$F(t) = \frac{1}{1 + \left(\frac{\max_{0 \leq \tau \leq t} D(\tau)}{m} \right)^{-\beta}}$ $S_{IT}(t) = (1 - F(t)) \times e^{-h_b \times t}$

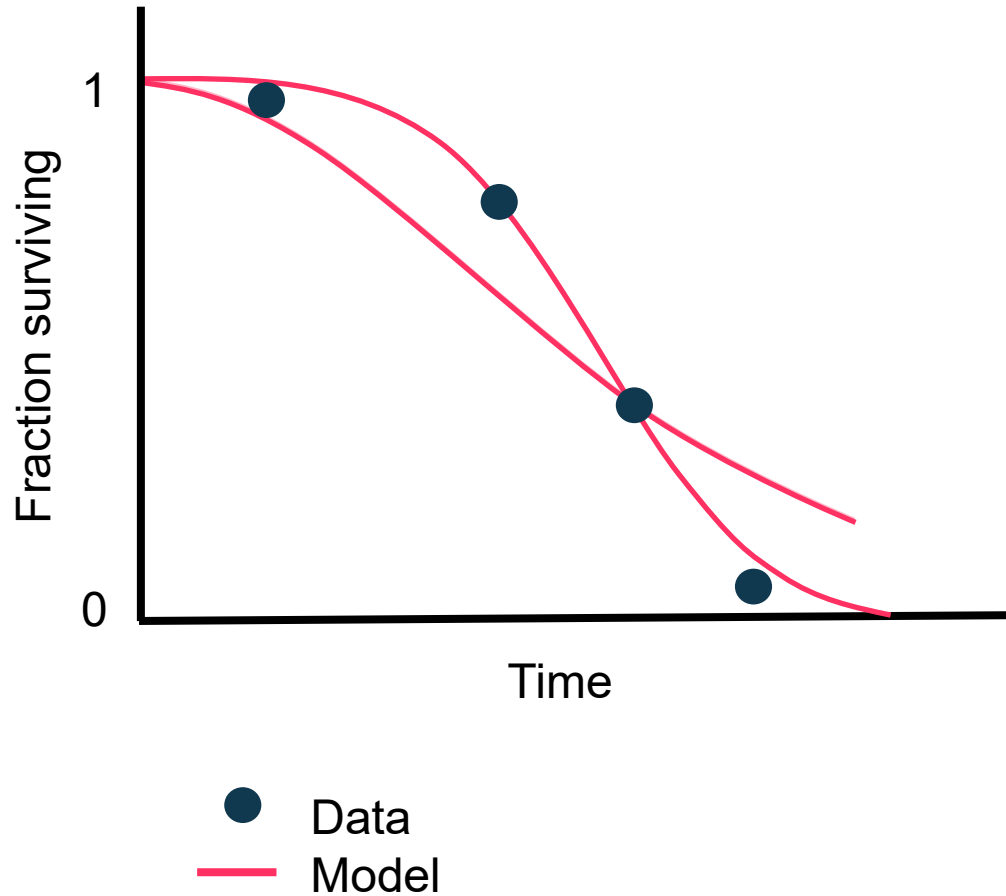


Exercise 6

- // Open a copy of your script from Exercise 5, delete all parts you have implemented to compare different model variants, keep the one-parameter TK function, and code for inputs, parameter structure, output collection and plotting.
- // Implement the *stochastic death* toxicodynamic assumption within the presented TK-TD model by using the *scaled damage* as dose metric (GUTS RED, slide 17). Therefore, calculate the hazard rate and survival as function of time. Parameter values need to be estimated in the next exercises.
- // Use constant exposure scenario and try different parameter values for k_d , b and z – does the model behave as expected?
- // Try different exposure concentrations – does the model behave as expected?

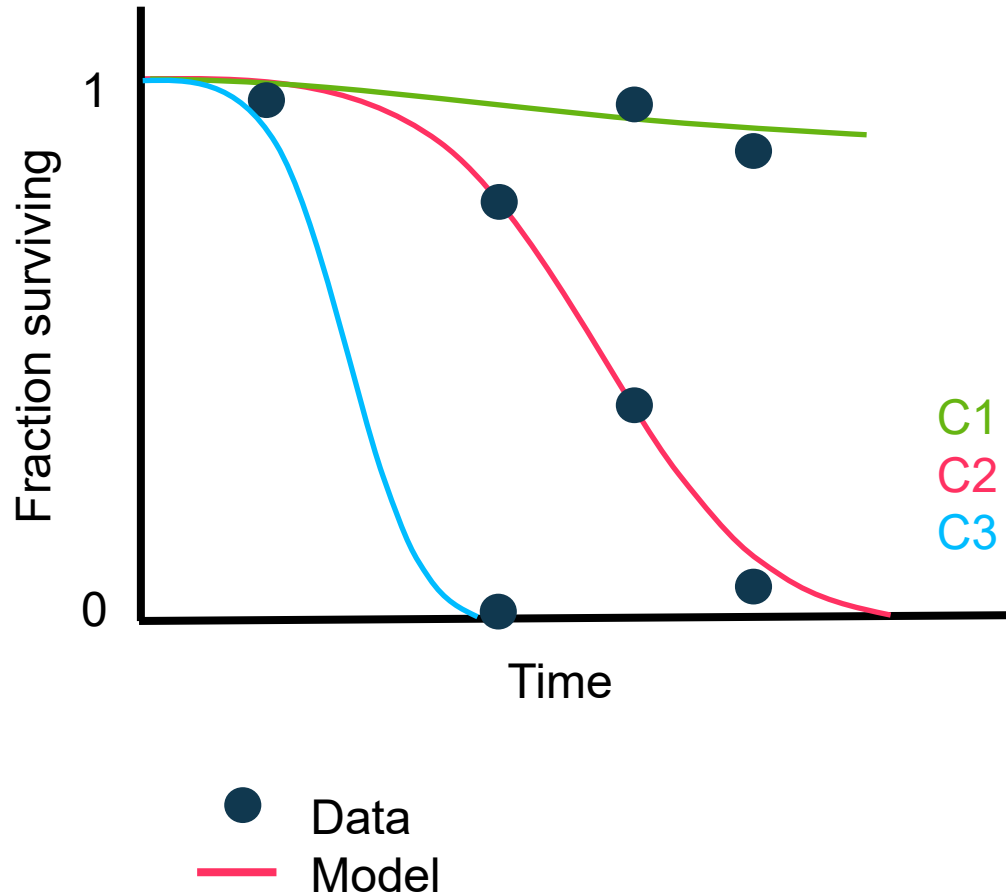
Some words on data requirements for GUTS parameter estimation

Data requirements for (reduced) GUTS parameter estimation



// Good idea on time course of the effect is needed (several data points in time)

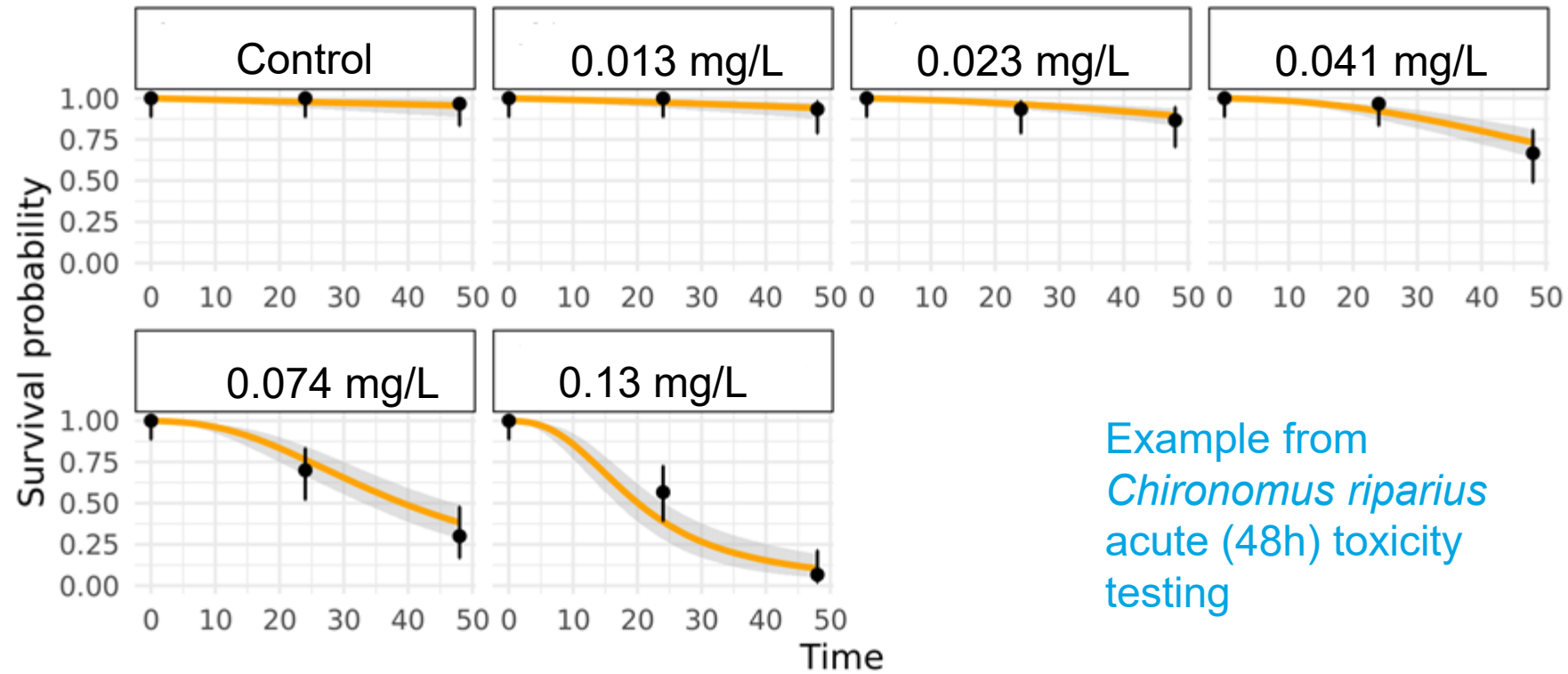
Data requirements for (reduced) GUTS parameter estimation



// Good idea on time course of the effect is needed (several data points in time)

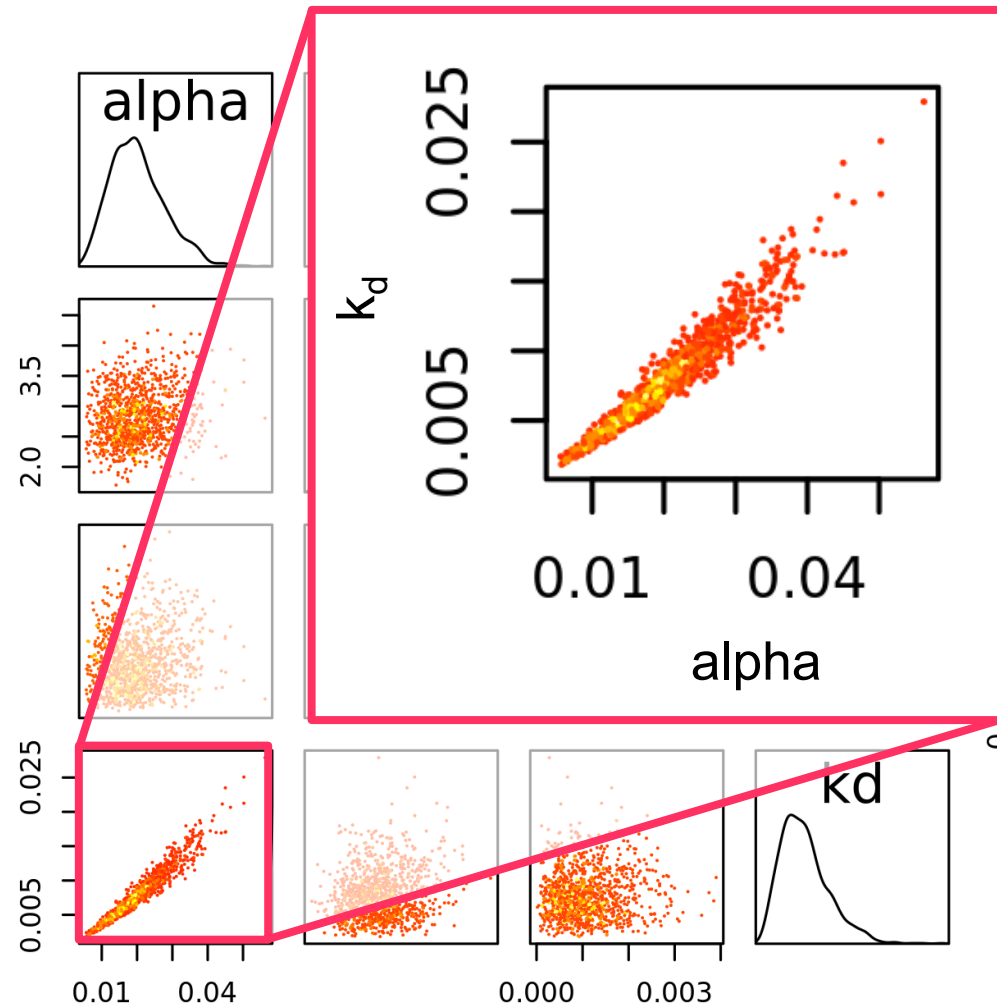
// Good information about concentration response is needed for estimation of threshold parameters and the slope components (+background mortality)

Example: GUTS-IT fit to toxicity data

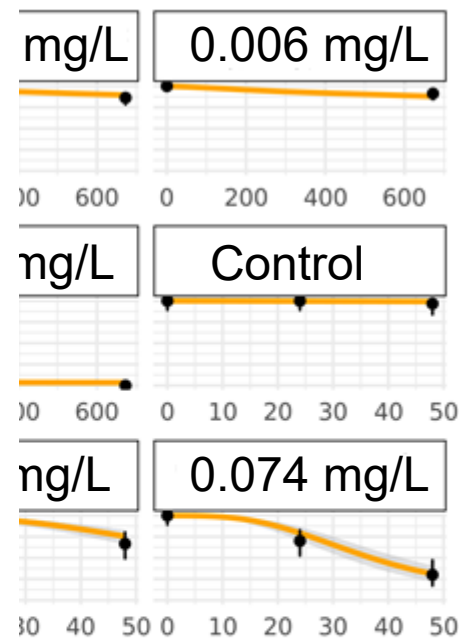
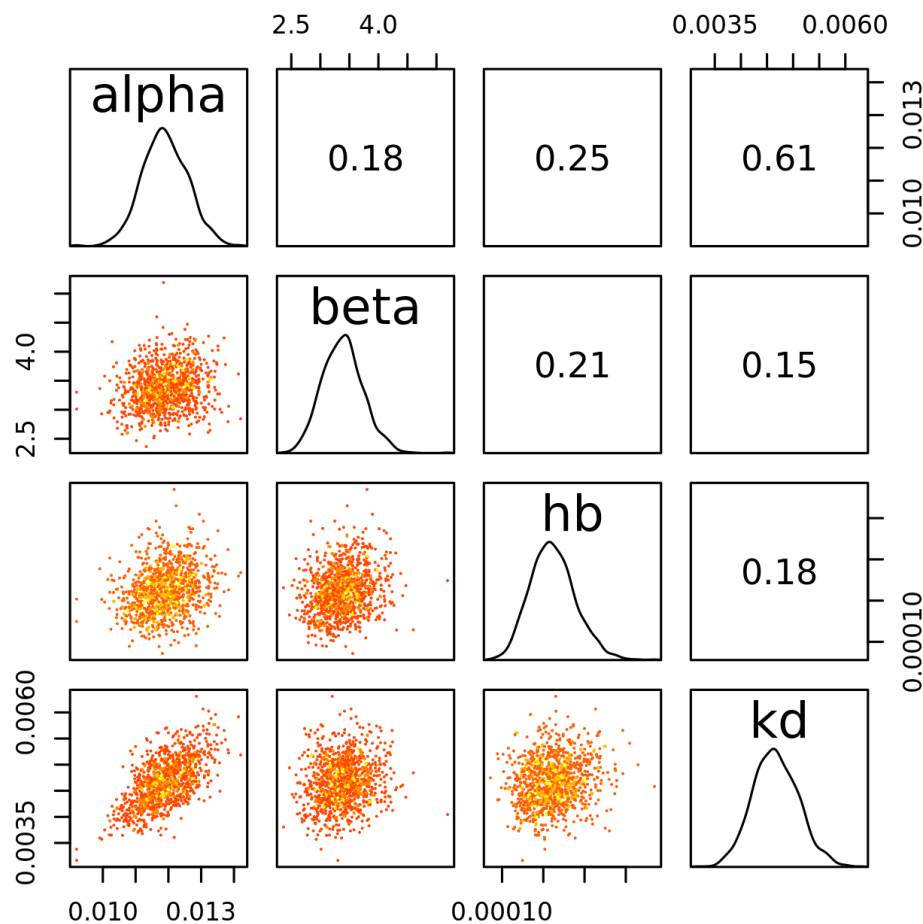


Example from
Chironomus riparius
acute (48h) toxicity
testing

GUTS-IT parameter correlations in this example

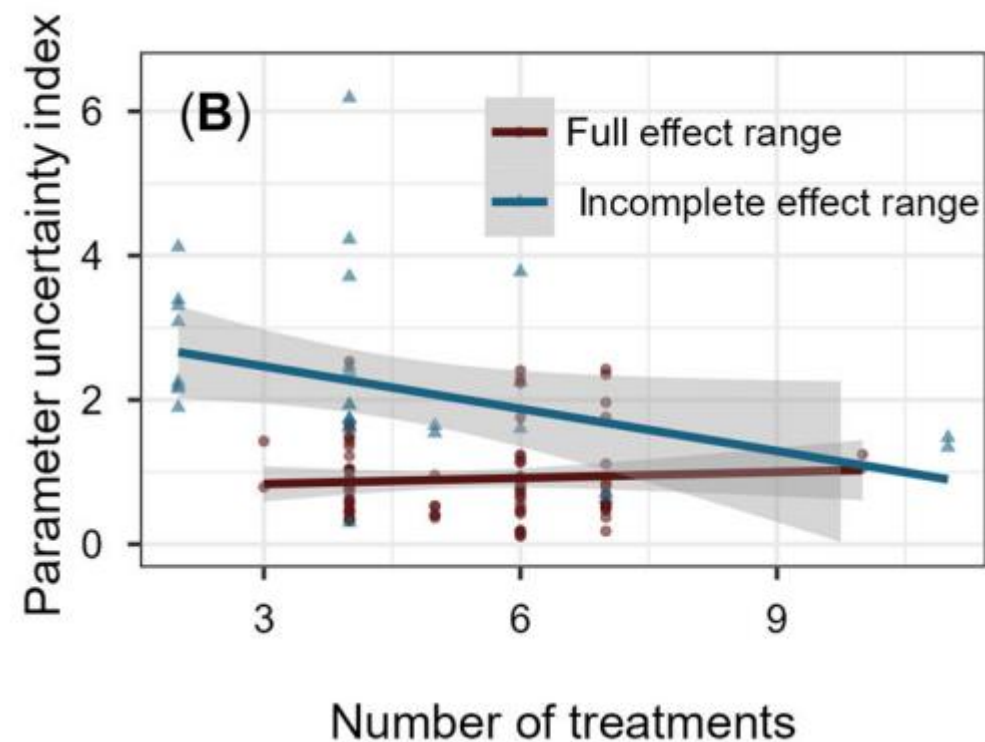
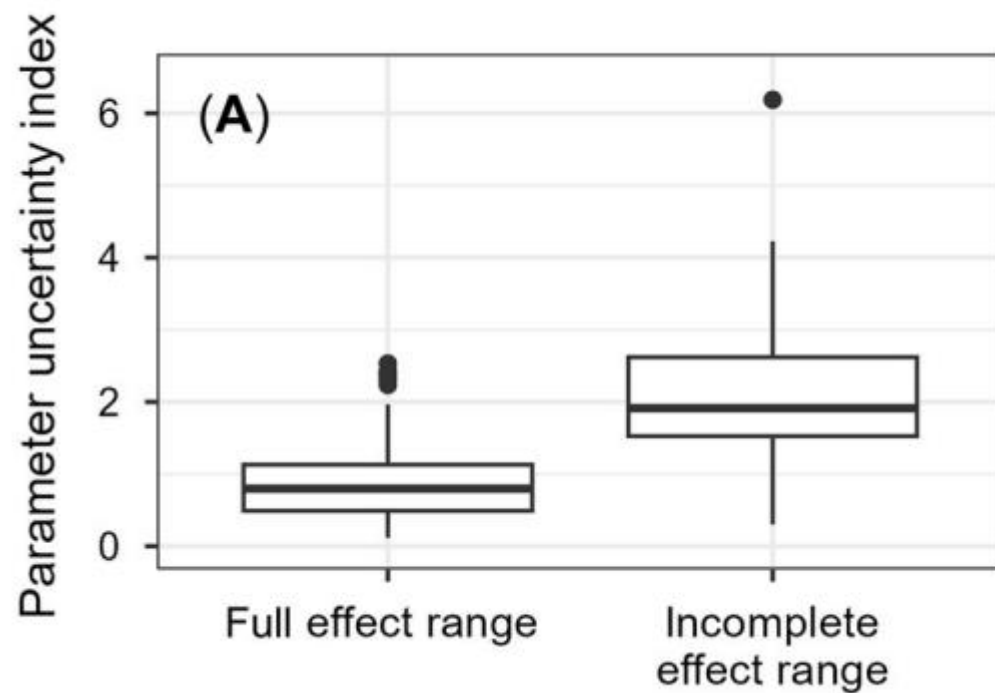


Example: GUTS-IT fit to more toxicity data



Example from *Chironomus riparius* acute & chronic toxicity testing

So, what is a good data set for GUTS modelling?

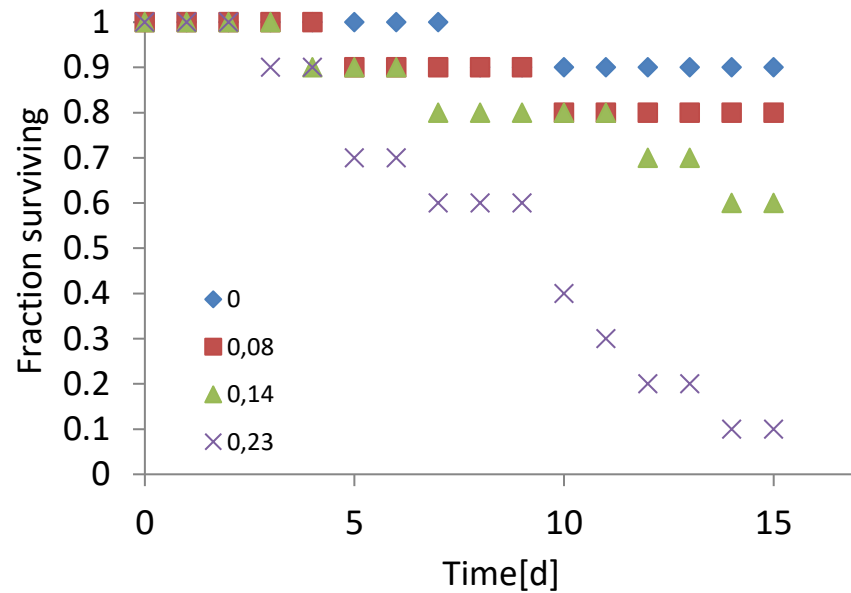


For further read: Bauer et al.
<https://doi.org/10.1002/etc.5761>



*Continue with
exercises*

Data set for exercise



In a 15 day experiment, survival of *Daphnia magna* was quantified for a control (no exposure) and three different cadmium exposure concentrations (0.08, 0.14, 0.23 mg/L)

For further information see Jager_etal 2006.pdf



Exercise 7

- // Parameterize the reduced GUTS for the toxicodynamic assumption of stochastic death based on the survival data using the `morse` package in R (prepared script: GUTS_Morse.R)
- // For parametrization, use the curated the survival data for *D. magna* and cadmium to meet the package requirements (also, see example in GUTS_Morse.R script)

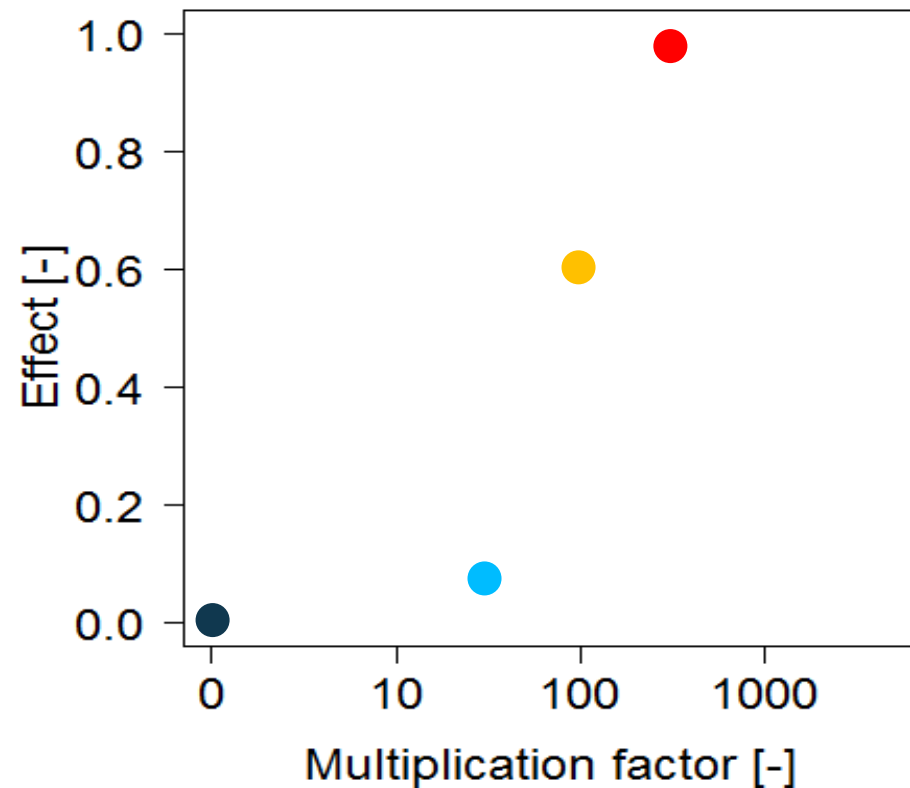
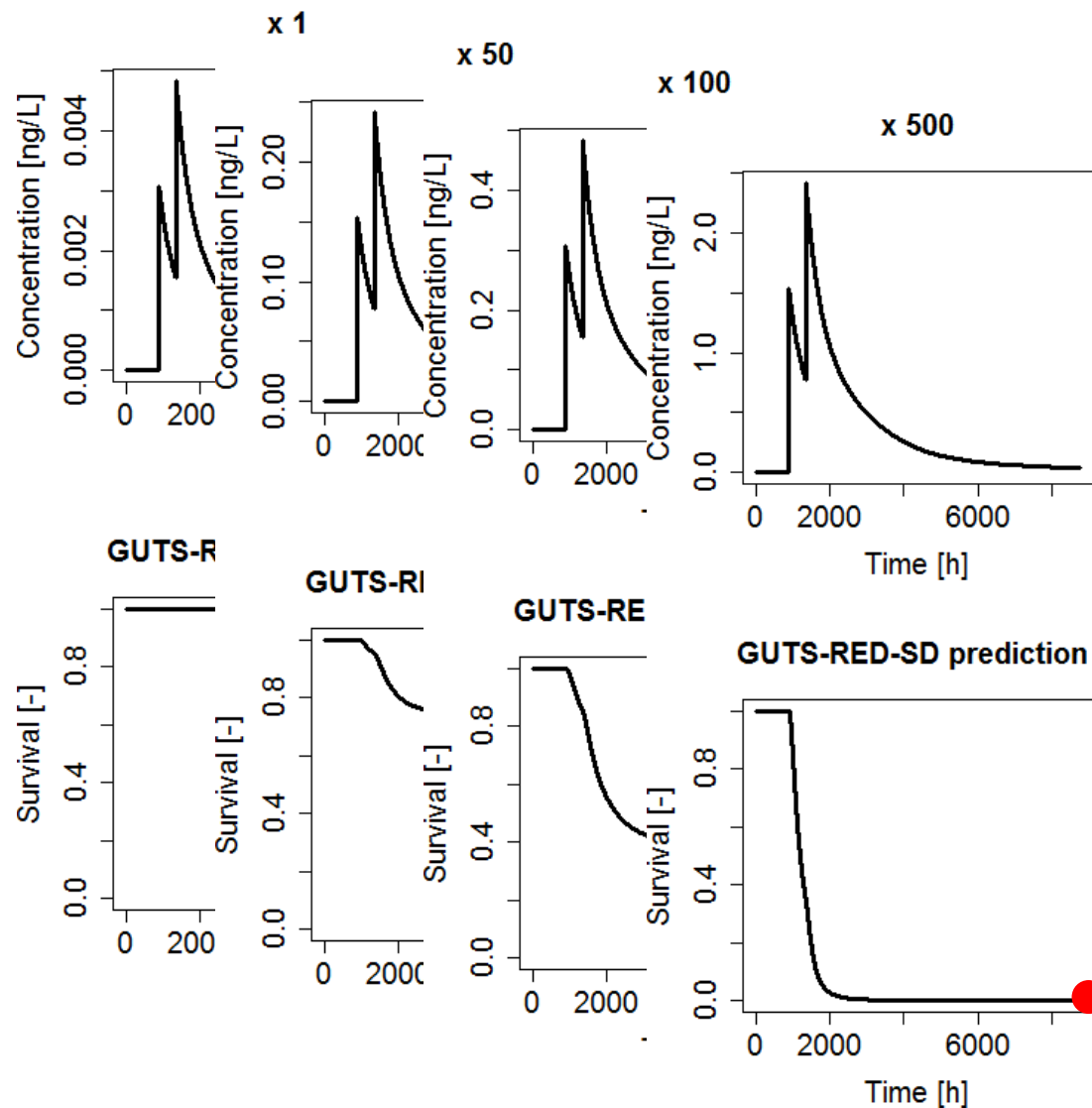


Exercise 8

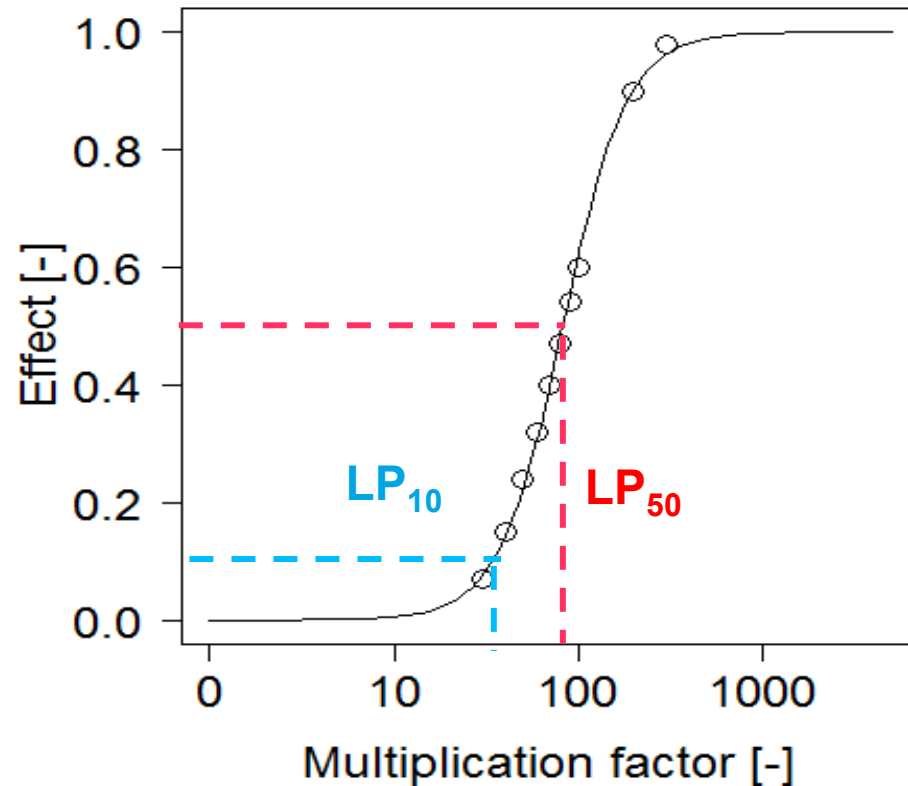
// Based on the script from Exercise 6 and the parameter values estimated in Exercise 7 (but $hb = 0$), simulate the effects for the two exposure profiles used in Exercise 2. What do you observe, and why?

GUTS in risk assessment: calculation of LPx values

Effect simulations for exposure scenario and different multiplication factors



GUTS in risk assessment: calculation of LP_x values



// Effect simulations for exposure scenario

// LP_x is the multiplication factor that is needed to cause x% effect for a given exposure scenario

// LP_x is calculated for all of the relevant exposure scenarios

// LP_x values can be compared with regular assessment factors



Exercise 9

// Calculate LP50 values for the two exposure profiles, therefore:

1) Calculate the effect at the end of the exposure period for a range of multiplication factors.

Considerations:

- a) How to calculate the effect?
- b) What is a meaningful range for the multiplication factors?
- c) Create a loop for the calculations.
- d) Conceive a way to store the results.

2) Estimate LP50 values using the drc package in R:

- a) plot results
- b) interpret DRC statistics

// Do you observe any differences between the exposure profiles? Why?